Module 1: Iterators (X Functions)

1.1 Concept

Iterator functions (`SUMX` , `AVERAGEX` , `MAXX` , `MINX`) evaluate an expression **row by row** across a table, then aggregate the results.

👉 Unlike `SUM(Sales[SalesAmount])` which just sums a column, `SUMX` lets you recompute values dynamically (e.g., `Quantity * UnitPrice * (1 - Discount)`), and then sum them.

1.2 Key Functions Overview

Function	What it Does	Example (on dataset)
`SUMX`	Iterates through rows and sums the expression	`SUMX(Sales, Sales[Quantity] * Sales[UnitPrice] * (1 - Sales[Discount]))`
`AVERAGEX`	Averages the result of row-by-row calculation	`AVERAGEX(Sales, Sales[Profit])`
`MAXX`	Returns maximum value across rows	`MAXX(Sales, Sales[Profit])`
`MINX`	Returns minimum value across rows	`MINX(Sales, Sales[Profit])`

1.3 Practical Examples

Example 1: Recalculate Total Sales

```
Total Sales :=
SUMX(
    Sales,
    Sales[Quantity] * Sales[UnitPrice] * (1 - Sales[Discount])
)
```

- Cross-checks the given `SalesAmount` column by recalculating.

Example 2: Average Profit per Transaction

```
Average Profit :=
AVERAGEX(Sales, Sales[Profit])
```

- Gives mean profit across all transactions.

Example 3: Maximum Profit on a Single Sale

```
Max Profit per Sale :=
MAXX(Sales, Sales[Profit])
```

- Highest recorded single-sale profit.

Example 4: Minimum Profit on a Single Sale

```
Min Profit per Sale :=
MINX(Sales, Sales[Profit])
```

- Lowest recorded single-sale profit.

1.4 Exercises (Hands-On)

1. **Recalculate Profit Margin %**
   Create a new measure:

   ```
   Profit Margin % :=
   AVERAGEX(
       Sales,
       DIVIDE(Sales[Profit], Sales[SalesAmount], 0)
   )
   ```

   👉 Average row-level profit margin instead of overall margin.
2. **Top vs Bottom Sales Amount**
   - Create `Top Sales Amount` using:

     ```
     MAXX(Sales, Sales[SalesAmount])
     ```

   - Create `Lowest Sales Amount` using:

     ```
     MINX(Sales, Sales[SalesAmount])
     ```

3. **Category-wise Average Sales**

   ```
   Avg Sales by Category :=
   AVERAGEX(
       Sales,
       Sales[SalesAmount]
   )
   ```

   👉 Place in a table visual with **Category** to compare.

1.5 Visualizations in Power BI

1. **KPI Cards**
   - Total Sales (SUMX)
   - Average Profit (AVERAGEX)
   - Max Profit (MAXX)
   - Min Profit (MINX)
2. **Table/Matrix**
   - Category | Product | Total Sales (SUMX) | Avg Profit (AVERAGEX)
3. **Bar Chart**
   - X-axis = Product
   - Y-axis = Total Sales (SUMX)

1.6 Exercise Answers (For Trainers)

1. **Profit Margin % (average row level)**
   - Likely range: **10–30%**, since dataset simulates profit = 10–30% of sales.
2. **Top vs Bottom Sales Amount**
   - Top Sale: Expect **close to 10,000+** (large unit price × multiple qty, no discount).
   - Lowest Sale: Could be **< 100** (small quantity, small unit price, with discount).
3. **Category-wise Average Sales**
   - Electronics will usually show **higher averages** due to products like Laptops & Phones.
   - Fashion tends to show **lower averages** (Shoes, T-Shirts).
   - Accessories somewhere in between.

(Exact numbers will depend on the dataset version but trends will remain consistent.)

✅ By the end of Module 1, students will:

- Understand **iterators vs aggregators**.
- Recalculate sales and profit row by row.
- Apply `SUMX`, `AVERAGEX`, `MAXX`, `MINX` to real dataset scenarios.